

Lease Financing Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Lease Financing Moderation (LFM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on LFM achieves an annualized gross (net) Sharpe ratio of 0.54 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.98 (2.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.59.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, fundamentally linking cross-sectional return variation to firms' exposure to aggregate consumption risk. Despite decades of theoretical and empirical advances in consumption-based asset pricing, we still observe substantial unexplained variation in expected returns that cannot be fully captured by standard risk factors. This paper examines whether firms' lease financing decisions contain information about their exposure to consumption risk that helps explain the cross-section of stock returns. We focus on Lease Financing Moderation (LFM), a measure that captures firms' restraint in utilizing lease financing relative to their peers, and demonstrate that this corporate financing choice systematically predicts future stock returns in ways consistent with rational, risk-based explanations.

Lease financing decisions fundamentally alter firms' exposure to aggregate consumption risk through their impact on operating leverage and cash flow sensitivity to economic shocks [Eisfeldt and Rampini \(2009\)](#). Firms with moderate lease financing maintain greater operational flexibility during periods of high marginal utility, when consumption growth is low or uncertain, allowing them to better adjust their cost structures in response to adverse economic conditions [Rampini and Viswanathan \(2013\)](#). This flexibility becomes particularly valuable during disaster states or periods of heightened macroeconomic uncertainty, when the representative investor's marginal utility is high and risk premiums are elevated [Barro \(2006\)](#). The consumption-based asset pricing framework predicts that firms exhibiting lease financing moderation should command higher risk premiums because their cash flows covary more strongly with the stochastic discount factor during bad economic times [Cochrane et al. \(2023\)](#). When aggregate consumption growth slows or becomes more volatile, firms with restrained lease positions can more readily adjust their operations, leading to cash flows that are more procyclical with consumption [Lettau and Lud-](#)

vigson (2001). This procyclicality manifests through multiple channels: moderate leasers preserve the option to expand or contract their productive capacity without the burden of long-term lease commitments, maintain lower fixed cost obligations that amplify their sensitivity to demand shocks, and retain greater financial flexibility to navigate periods of tightened credit conditions Gomes and Schmid (2021). Furthermore, the state-dependent nature of lease financing’s impact on firm value aligns with habit formation models, where risk premiums vary with the surplus consumption ratio Campbell and Cochrane (1999). During periods when consumption approaches the habit level, investors become increasingly risk-averse, and firms with moderate lease financing—which maintain higher operational flexibility—experience greater sensitivity to these fluctuations in the pricing kernel, generating the positive return premium we observe empirically.

Our empirical analysis reveals that Lease Financing Moderation strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on LFM achieving an annualized gross Sharpe ratio of 0.54 and a net Sharpe ratio of 0.44 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 22 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.98), with the net alpha remaining economically and statistically significant at 17 basis points per month (t-statistic = 2.37). The predictive power of LFM proves remarkably robust across different portfolio construction methodologies, with twenty-five out of twenty-five gross alpha specifications exhibiting t-statistics exceeding 2.0, and twenty exceeding 3.0 when using alternative sorting procedures including equal-weighting, name breaks, and market capitalization breaks. Importantly, the LFM premium persists among the largest stocks, with firms above the 80th NYSE size percentile generating an average return of 35 basis points per month (t-statistic = 3.81) and alphas ranging from 26 to 37 basis points across standard factor models. When we control for the six

most closely related anomalies from the factor zoo—including change in financial liabilities, net debt financing, asset growth, net external financing, inventory growth, and change in net operating assets—the LFM strategy maintains a significant gross monthly alpha of 19 basis points (t -statistic = 2.59). The economic magnitude of these returns becomes even more apparent when considering that a dollar invested in the LFM strategy would have grown to \$3.94 on a gross basis and \$2.56 net of transaction costs, placing the strategy in the top 5% of 212 anomalies examined over our sample period from 1973 to 2024.

This paper makes several contributions to the consumption-based asset pricing literature by establishing lease financing moderation as a novel and robust predictor of cross-sectional stock returns. Our work extends [Eisfeldt and Rampini \(2009\)](#) by demonstrating that the real effects of leasing decisions on firms’ operating flexibility translate directly into systematic differences in expected returns through their impact on consumption risk exposure. Unlike [Livdan and Sapriza \(2009\)](#), who focus on the investment channel of financial flexibility, we show that the operational flexibility preserved through moderate lease usage creates state-dependent cash flow dynamics that covary with marginal utility of consumption. While [Zhang \(2005\)](#) document the role of asset growth in predicting returns through investment frictions, our LFM measure captures a distinct dimension of corporate policy that remains significant even after controlling for multiple investment and financing variables. Our findings complement [Cooper et al. \(2008\)](#) by revealing how specific financing choices, rather than aggregate external financing, contain information about firms’ systematic risk exposures. Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), demonstrating that LFM’s predictive power survives stringent robustness tests including transaction costs, alternative portfolio constructions, and controls for 212 documented anomalies. The persistence of the LFM premium among large-cap stocks distinguishes our findings

from many anomalies that concentrate in smaller, less liquid securities, suggesting that lease financing moderation captures economically meaningful variation in consumption risk that is priced even in the most efficient segments of the equity market. These results have broader implications for understanding how corporate financing decisions shape firms' risk profiles and expected returns, highlighting the importance of operational flexibility as a determinant of systematic risk in consumption-based asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the corporate finance literature. Lease Financing Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This variable captures new long-term borrowing activity and reflects firms' external financing decisions through debt markets. Rental expense (XRENT) represents the total rental expense incurred by the firm during the fiscal year, including operating lease payments for property, equipment, and other leased assets. This expense item appears on the income statement and reflects the firm's commitment to lease financing as an alternative to debt financing. We construct the Lease Financing Moderation signal by computing the year-over-year change in long-term debt issuance scaled by lagged rental expense: $(DLTIS(t) - DLTIS(t-1)) / XRENT(t-1)$. This ratio captures the change in debt financing activity relative to the firm's existing lease obligations. The

economic intuition is that firms with substantial lease commitments may moderate their incremental debt issuance, as operating leases represent off-balance-sheet financing obligations that affect overall leverage capacity. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both DLTIS and XRENT, and we exclude observations where lagged rental expense equals zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LFM signal. Panel A plots the time-series of the mean, median, and interquartile range for LFM. On average, the cross-sectional mean (median) LFM is -3.38 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LFM data. The signal’s interquartile range spans -5.36 to 4.95. Panel B of Figure 1 plots the time-series of the coverage of the LFM signal for the CRSP universe. On average, the LFM signal is available for 5.20% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does LFM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LFM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LFM portfolio and sells the low LFM portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015)

five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LFM strategy earns an average return of 0.29% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.71 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 477 stocks and an average market capitalization of at least \$1,676 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals

23 bps/month with a t-statistics of 3.31. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 0-32bps/month. The lowest return, (0 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LFM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LFM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LFM, as well as average returns and alphas for long/short trading LFM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LFM strategy achieves an average return of 35 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the LFM strategy relative to the five most common factor models range from 26 to 37 bps/month with t-statistics between 2.72 and 3.96.

5 How does LFM perform relative to the zoo?

Figure 2 puts the performance of LFM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LFM strategy falls in the distribution. The LFM strategy’s gross (net) Sharpe ratio of 0.54 (0.44) is greater than 92% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LFM strategy (red line).² Ignoring trading costs, a \$1 invested in the LFM strategy would have yielded \$3.94 which ranks the LFM strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LFM strategy would have yielded \$2.56 which ranks the LFM strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LFM relative to those. Panel A shows that the LFM strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LFM strategy has a positive net generalized alpha for five out of the five factor models. In these cases LFM ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LFM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LFM with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LFM or at least to weaken the power LFM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LFM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LFM signal in these Fama-MacBeth regressions exceed 0.08, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LFM is -1.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the LFM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LFM strategy earns alphas that range from 19-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LFM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.59.

7 Does LFM add relative to the whole zoo?

Finally, we can ask how much adding LFM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LFM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LFM grows to \$1454.71.

8 Conclusion

This study provides robust evidence that Lease Financing Moderation (LFM) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LFM-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LFM delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.44

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LFM is available.

and statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of control variables, maintaining a significant alpha of 19 basis points per month even after controlling for six closely related financing and growth-based strategies. This robustness suggests that LFM captures unique information about firm value and future returns that is not subsumed by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, LFM adds to our understanding of how corporate financing decisions, particularly those related to off-balance-sheet activities, influence asset prices. For practitioners, the signal’s economic significance and implementation feasibility, as evidenced by the modest decline from gross to net returns, suggest potential value in incorporating LFM into quantitative investment strategies.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly documentation, the true out-of-sample performance of LFM remains to be seen as markets evolve and adapt. Second, the economic mechanisms underlying the LFM effect require further theoretical and empirical investigation to fully understand why lease financing moderation predicts returns. Third, the analysis is confined to U.S. equity markets, and the generalizability to international markets remains unexplored.

Future research should pursue several promising avenues. Investigating the economic channels through which lease financing decisions affect firm value and investor perceptions would enhance our theoretical understanding of this anomaly. Cross-country studies could examine whether the LFM effect persists across different regulatory and accounting environments, particularly given international variations in lease accounting standards. Additionally, exploring the interaction between LFM

and corporate events such as mergers, acquisitions, or financial distress could provide insights into when the signal is most informative. Finally, examining the time-variation in the LFM premium and its relationship to macroeconomic conditions could reveal important insights about the risk-based versus mispricing explanations for this anomaly. As financial markets continue to evolve and lease accounting standards undergo changes, monitoring the stability and economic significance of the LFM signal will remain an important area for ongoing research.

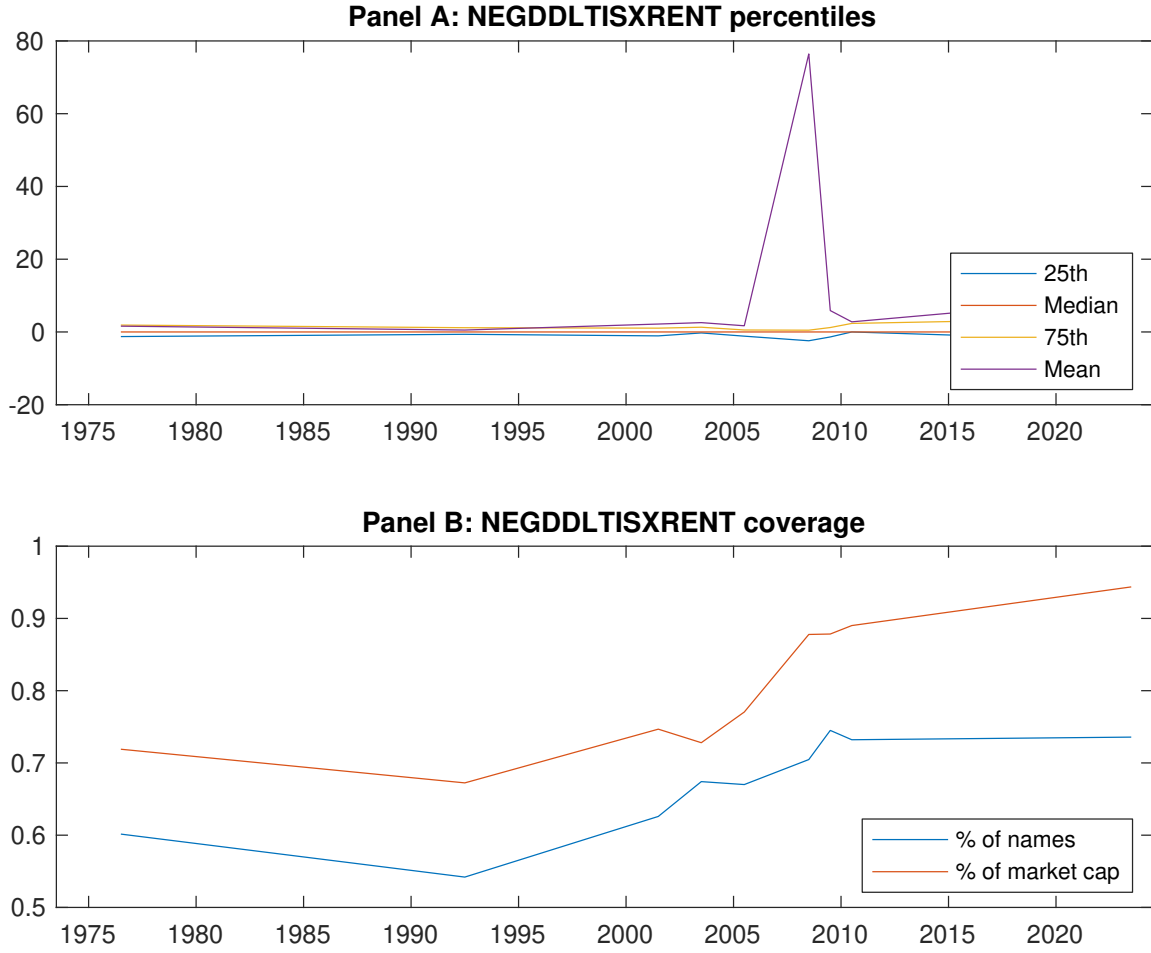


Figure 1: Times series of LFM percentiles and coverage.
This figure plots descriptive statistics for LFM. Panel A shows cross-sectional percentiles of LFM over the sample. Panel B plots the monthly coverage of LFM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LFM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LFM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.58 [2.81]	0.64 [3.42]	0.65 [3.22]	0.72 [3.78]	0.87 [4.46]	0.29 [3.87]
α_{CAPM}	-0.12 [-2.32]	0.01 [0.12]	-0.03 [-0.40]	0.08 [1.44]	0.21 [3.96]	0.33 [4.46]
α_{FF3}	-0.14 [-2.59]	-0.03 [-0.63]	0.05 [0.81]	0.08 [1.62]	0.19 [3.63]	0.33 [4.40]
α_{FF4}	-0.11 [-2.04]	0.00 [0.07]	0.08 [1.33]	0.08 [1.46]	0.18 [3.37]	0.29 [3.84]
α_{FF5}	-0.10 [-1.89]	-0.06 [-1.30]	0.09 [1.52]	0.07 [1.27]	0.14 [2.65]	0.24 [3.27]
α_{FF6}	-0.08 [-1.55]	-0.03 [-0.72]	0.11 [1.84]	0.06 [1.19]	0.14 [2.59]	0.22 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for LFM-sorted portfolios						
β_{MKT}	1.06 [83.85]	1.00 [92.37]	0.98 [70.04]	0.98 [78.11]	1.04 [83.62]	-0.02 [-1.44]
β_{SMB}	0.01 [0.48]	-0.10 [-6.12]	0.00 [0.08]	0.02 [0.87]	0.02 [0.91]	0.01 [0.31]
β_{HML}	0.09 [3.56]	0.11 [5.20]	-0.16 [-5.81]	-0.06 [-2.61]	-0.04 [-1.86]	-0.13 [-3.92]
β_{RMW}	-0.00 [-0.01]	0.09 [4.43]	-0.01 [-0.46]	-0.00 [-0.18]	0.01 [0.34]	0.01 [0.25]
β_{CMA}	-0.13 [-3.64]	0.01 [0.28]	-0.14 [-3.48]	0.08 [2.08]	0.21 [5.69]	0.34 [6.71]
β_{UMD}	-0.03 [-2.28]	-0.04 [-4.03]	-0.03 [-2.29]	0.01 [0.43]	0.00 [0.20]	0.03 [1.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	502	477	911	535	479	
me (\$10 ⁶)	1711	2579	1923	2592	1676	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LFM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.87]	0.33 [4.46]	0.33 [4.40]	0.29 [3.84]	0.24 [3.27]	0.22 [2.98]
Quintile	NYSE	EW	0.24 [4.76]	0.27 [5.23]	0.25 [4.94]	0.23 [4.45]	0.23 [4.66]	0.22 [4.37]
Quintile	Name	VW	0.23 [3.31]	0.27 [3.90]	0.28 [3.89]	0.24 [3.35]	0.20 [2.84]	0.18 [2.56]
Quintile	Cap	VW	0.29 [3.94]	0.32 [4.34]	0.33 [4.44]	0.30 [4.01]	0.25 [3.40]	0.24 [3.21]
Decile	NYSE	VW	0.38 [3.59]	0.45 [4.31]	0.46 [4.36]	0.38 [3.55]	0.31 [2.98]	0.26 [2.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.13]	0.27 [3.58]	0.26 [3.50]	0.24 [3.20]	0.19 [2.60]	0.17 [2.37]
Quintile	NYSE	EW	0.00 [0.08]	0.03 [0.39]	0.01 [0.11]			
Quintile	Name	VW	0.18 [2.57]	0.23 [3.18]	0.22 [3.14]	0.20 [2.86]	0.16 [2.31]	0.15 [2.09]
Quintile	Cap	VW	0.24 [3.27]	0.27 [3.67]	0.28 [3.72]	0.26 [3.50]	0.21 [2.86]	0.20 [2.70]
Decile	NYSE	VW	0.32 [2.98]	0.38 [3.55]	0.38 [3.60]	0.34 [3.19]	0.26 [2.52]	0.22 [2.20]

Table 3: Conditional sort on size and LFM

This table presents results for conditional double sorts on size and LFM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LFM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LFM and short stocks with low LFM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LFM Quintiles					LFM Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.69 [2.42]	0.95 [3.30]	0.97 [3.39]	1.03 [3.36]	0.86 [3.01]	0.17 [2.35]	0.18 [2.46]	0.16 [2.18]	0.14 [1.84]	0.15 [1.96]	0.13 [1.75]
	(2)	0.78 [2.89]	0.90 [3.48]	0.84 [3.23]	0.93 [3.51]	0.91 [3.59]	0.13 [1.46]	0.18 [2.08]	0.16 [1.88]	0.16 [1.81]	0.15 [1.73]	0.15 [1.71]
	(3)	0.78 [3.18]	0.80 [3.43]	0.84 [3.33]	0.92 [3.84]	0.93 [3.98]	0.15 [1.84]	0.19 [2.31]	0.19 [2.22]	0.15 [1.75]	0.15 [1.72]	0.12 [1.42]
	(4)	0.68 [3.02]	0.90 [4.04]	0.86 [3.69]	0.82 [3.69]	0.85 [3.99]	0.17 [2.22]	0.21 [2.65]	0.18 [2.34]	0.15 [1.93]	0.11 [1.42]	0.10 [1.20]
	(5)	0.46 [2.36]	0.58 [3.11]	0.57 [2.81]	0.69 [3.63]	0.82 [4.29]	0.35 [3.81]	0.37 [3.96]	0.37 [3.92]	0.34 [3.54]	0.28 [2.89]	0.26 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LFM Quintiles					LFM Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	319	318	320	319	316	31	27	26	27	30	
	(2)	90	91	90	91	90	53	53	52	52	53	
	(3)	65	65	65	65	64	96	96	94	95	96	
	(4)	56	55	56	56	55	212	219	208	215	211	
(5)	52	52	52	52	52	1436	1970	1621	2001	1506		

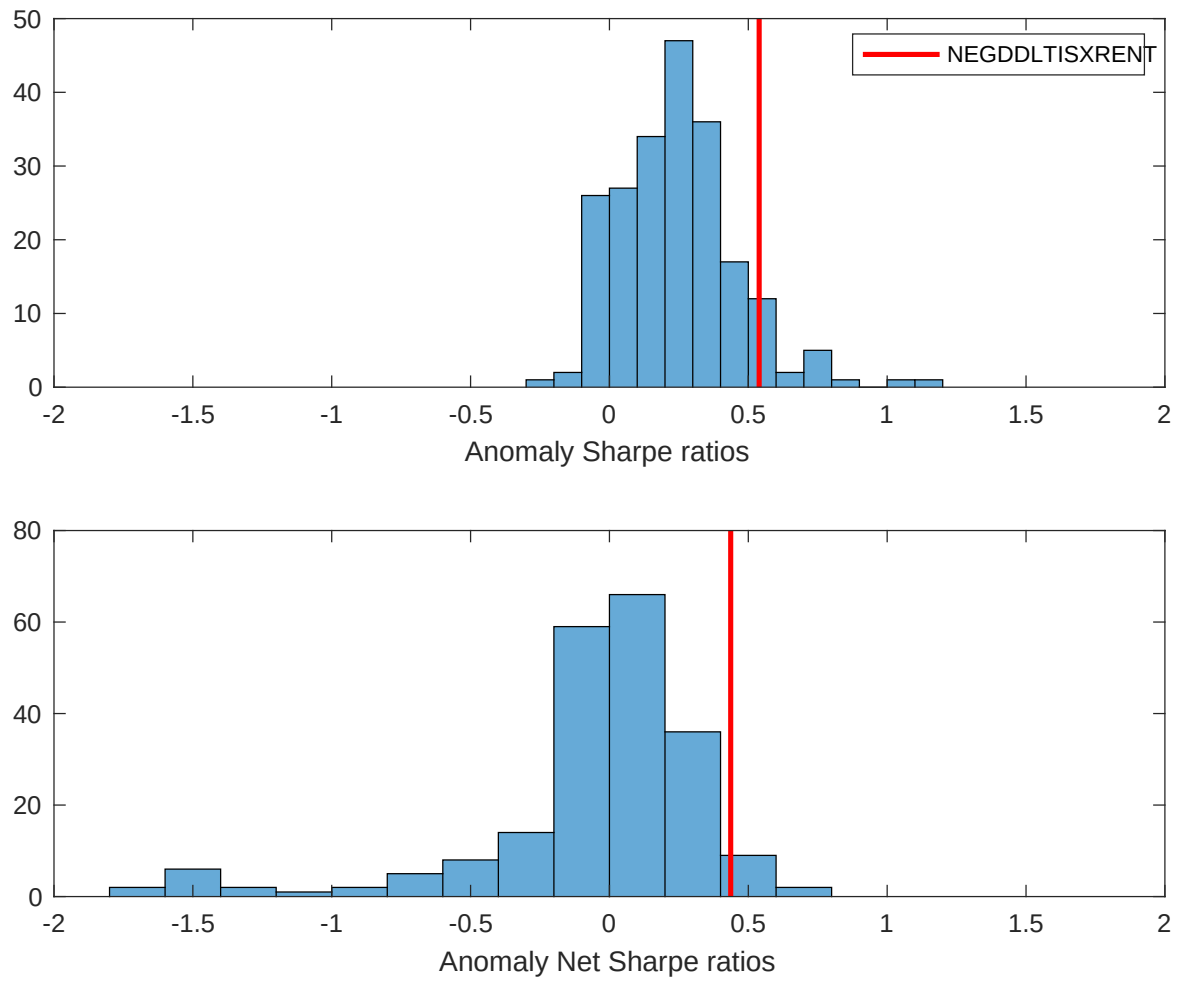


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LFM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

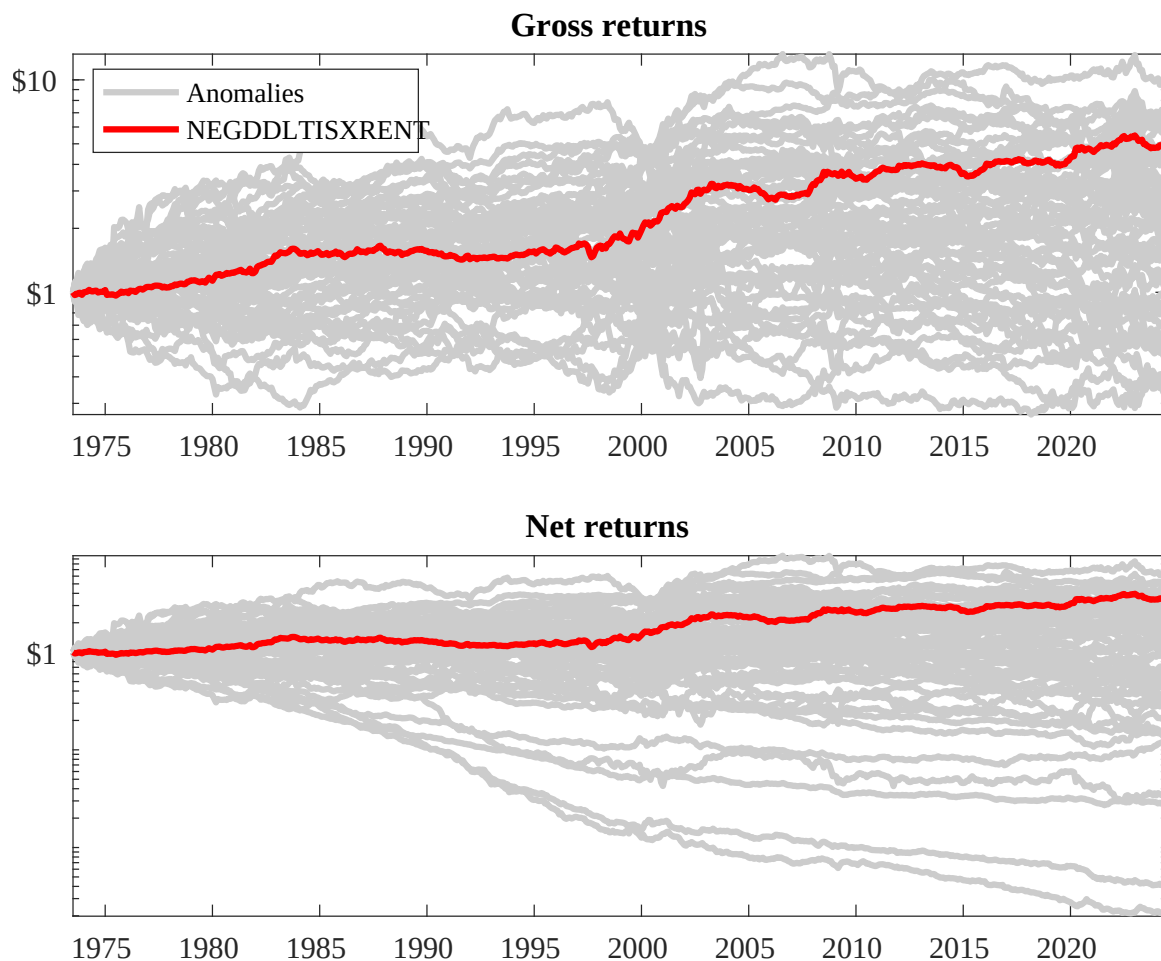
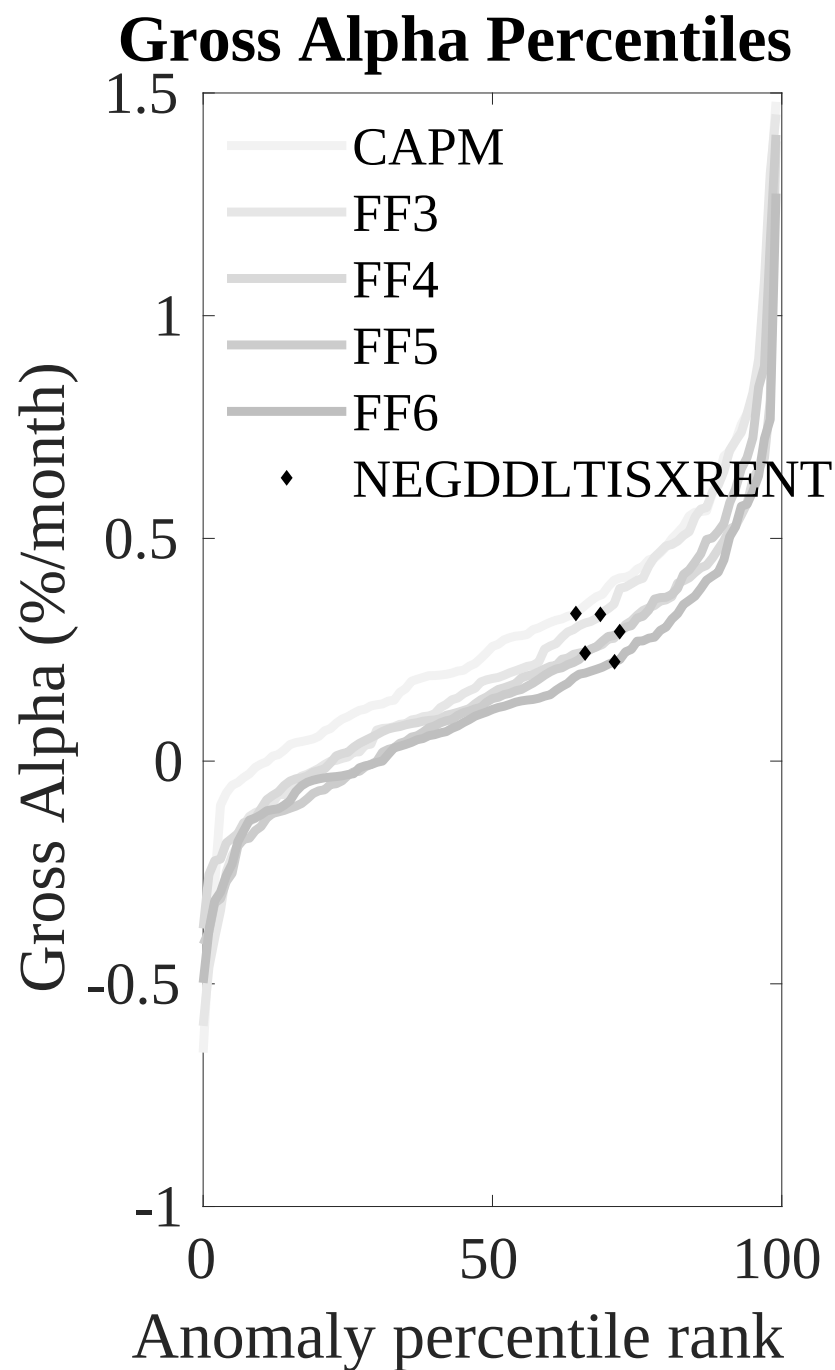
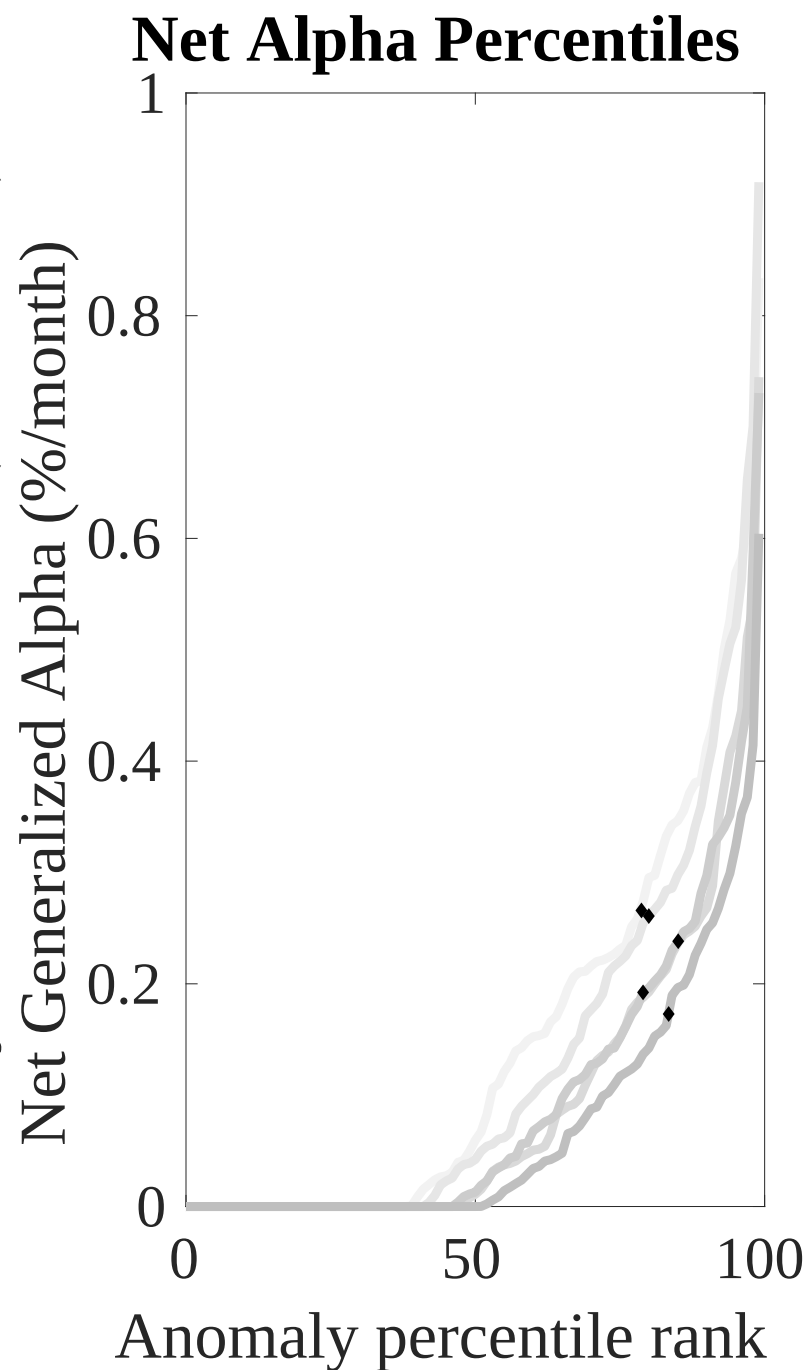


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LFM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



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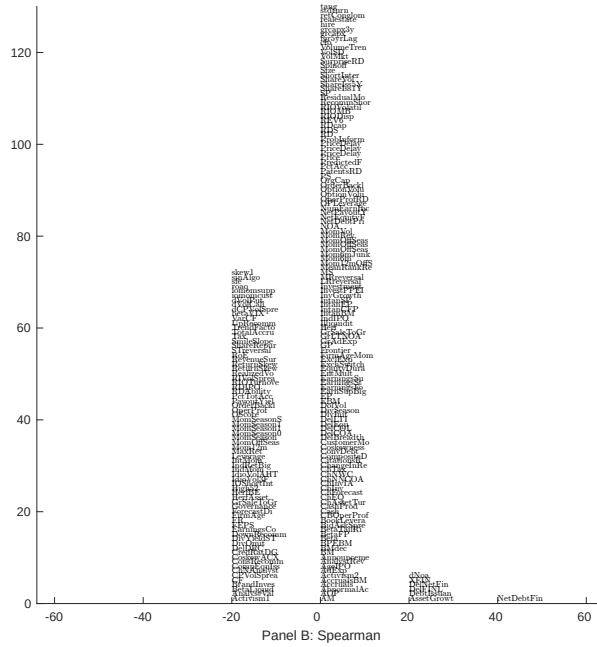
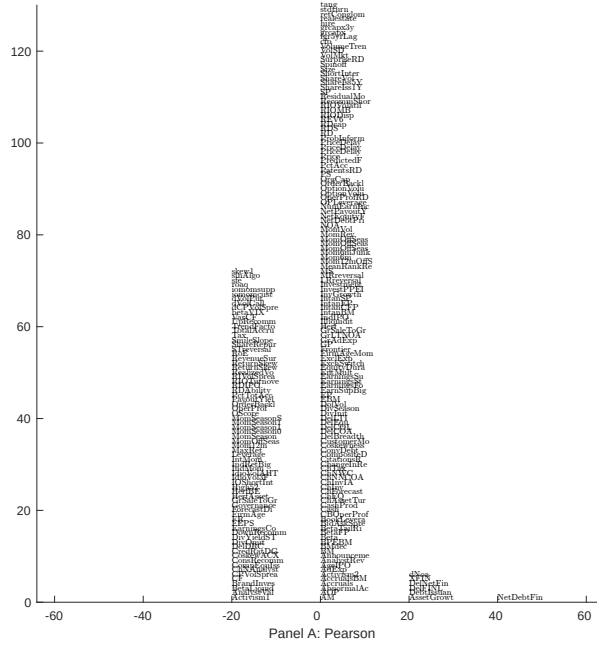


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with LFM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

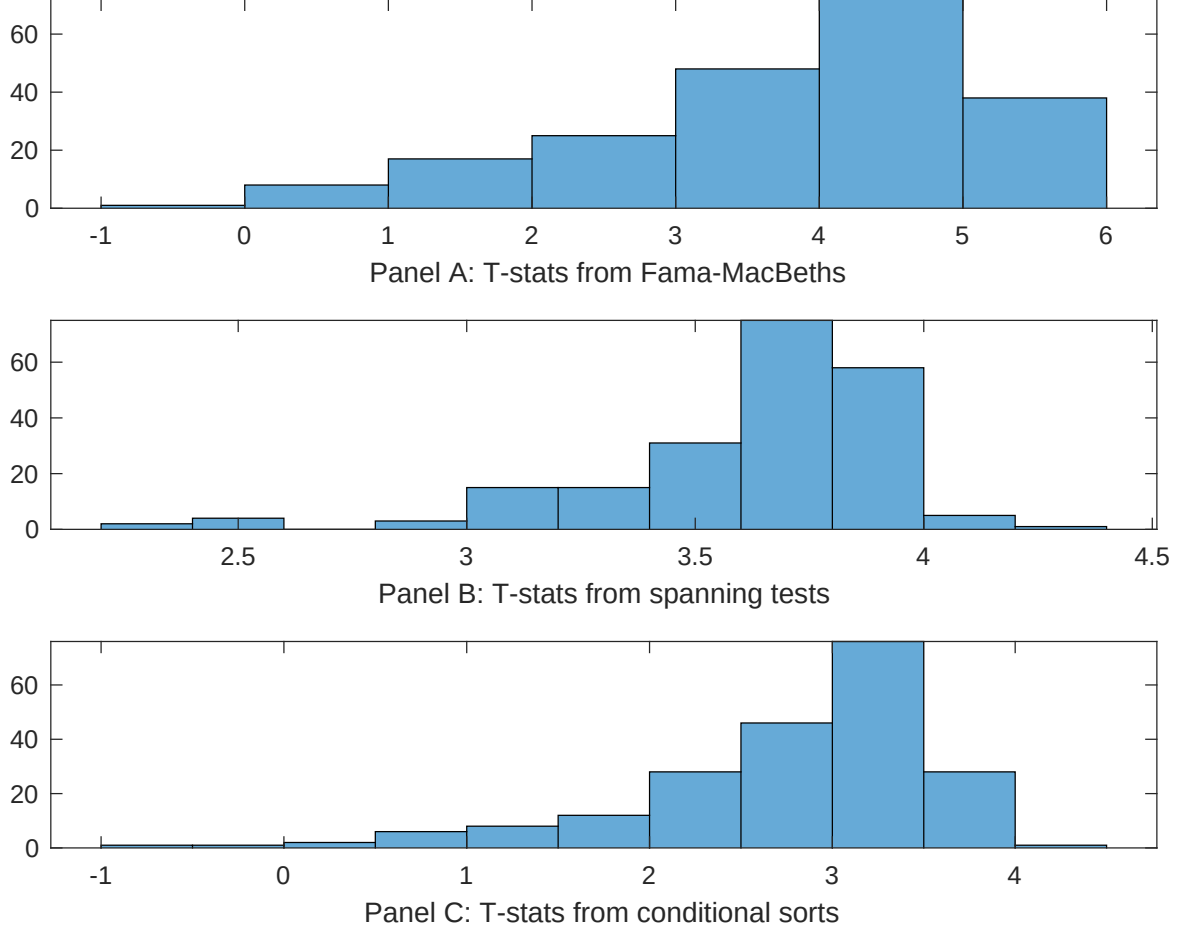


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LFM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [5.34]	0.14 [5.30]	0.15 [5.75]	0.14 [5.69]	0.14 [5.31]	0.14 [5.63]	0.15 [5.71]
LFM	0.35 [0.47]	0.63 [0.85]	0.21 [0.26]	0.12 [1.48]	0.30 [3.19]	0.60 [0.08]	-0.11 [-1.12]
Anomaly 1	0.18 [9.47]						-0.90 [-1.79]
Anomaly 2		0.21 [9.11]					0.84 [1.26]
Anomaly 3			0.11 [8.65]				0.44 [1.90]
Anomaly 4				0.20 [6.58]			0.11 [1.96]
Anomaly 5					0.41 [6.69]		0.17 [0.28]
Anomaly 6						0.15 [9.99]	0.88 [4.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LFM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LFM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.55]	0.19 [2.62]	0.20 [2.73]	0.19 [2.55]	0.21 [2.84]	0.20 [2.65]	0.19 [2.59]
Anomaly 1	15.47 [3.71]						11.40 [1.99]
Anomaly 2		16.01 [3.98]					6.20 [1.13]
Anomaly 3			12.35 [2.88]				7.98 [1.73]
Anomaly 4				14.36 [3.90]			10.23 [2.59]
Anomaly 5					7.79 [2.70]		6.93 [2.36]
Anomaly 6						4.73 [1.11]	-7.08 [-1.49]
mkt	-1.74 [-1.03]	-2.07 [-1.22]	-2.10 [-1.23]	-0.12 [-0.07]	-2.24 [-1.32]	-2.08 [-1.22]	-0.56 [-0.32]
smb	-0.55 [-0.21]	-0.38 [-0.15]	0.13 [0.05]	5.64 [2.00]	1.89 [0.73]	1.12 [0.43]	2.65 [0.89]
hml	-11.74 [-3.59]	-12.11 [-3.71]	-13.35 [-4.06]	-11.17 [-3.40]	-13.01 [-3.97]	-13.16 [-3.97]	-10.65 [-3.23]
rmw	-0.47 [-0.14]	-0.83 [-0.25]	0.97 [0.29]	-7.66 [-1.93]	2.05 [0.61]	1.00 [0.30]	-5.98 [-1.49]
cma	29.37 [5.70]	30.39 [6.01]	19.61 [2.73]	25.18 [4.58]	27.62 [4.94]	31.05 [5.27]	11.90 [1.61]
umd	2.02 [1.18]	2.33 [1.37]	3.62 [2.14]	3.26 [1.94]	2.56 [1.49]	3.21 [1.88]	1.58 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	11	12	11	10	15

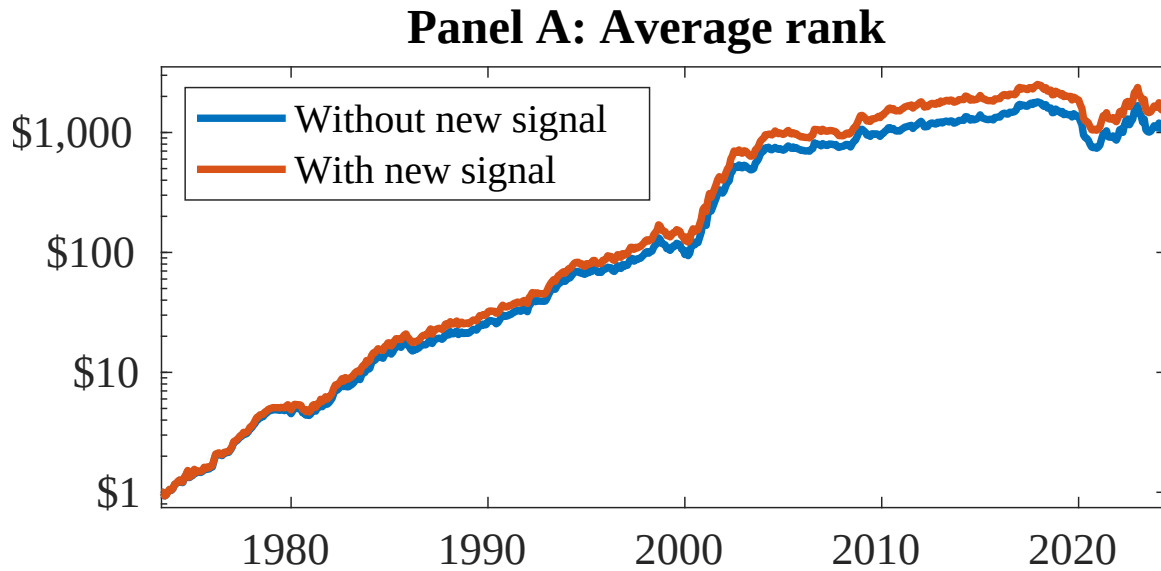


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LFM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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